



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

FUNDAMENTAL OF BLOCKCHAIN TECHNOLOGY [IT-4422]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Imagine a newly launched public blockchain network has rapidly gained popularity, but due to its small number of mining participants, one group of miners manages to gain control of more than 51% of the network's total hashing power. Apply your understanding of blockchain architecture and consensus mechanisms to explain how this situation could lead to a 51% attack. (5)
 - A)
 - B)
- 2) A multi logistics company is facing issues with data inconsistency and lack of transparency across its global supply chain partners. As a technology consultant, how would you apply Distributed Ledger Technology (DLT) to resolve these issues? (5)
 - A) Explain how DLT can improve data integrity, traceability, and trust among stakeholders.
 - B) Apply the concept of Zero-Knowledge Proofs (ZKPs) to design a privacy-preserving authentication system. Explain how ZKPs ensure user data confidentiality while maintaining verification accuracy. (5)
- 3) A global supply chain company is implementing smart contracts to automate payments between suppliers and distributors. The contract automatically executes payment when goods are received and verified through IoT sensors. Explain how smart contracts can help to reduce fraud and inefficiencies in the supply chain? (5)
 - A)
 - B) A hospital network wants to use smart contracts to securely share patient medical records between authorized doctors, insurance companies, and research institutions. Patients have control over who can access their data. Apply the concept of oracles to explain their role in integrating external health data into smart contract systems. (5)
- 4) Imagine you are building a decentralized content-sharing platform that avoids reliance on centralized servers. As a blockchain developer, how would you leverage the key technologies of the InterPlanetary File System (IPFS) to design a scalable and secure system for storing and retrieving large media files? (5)
 - A)
 - B) Apply your understanding of NFTs to analyse real-world use cases such as digital art, gaming assets, music rights. (5)
- 5) A decentralized finance (DeFi) startup is exploring platforms to integrate a decentralized prediction market into its ecosystem. The team is considering using (5)

- A) Augur to allow users to bet on the outcomes of events ranging from elections to sports and crypto prices. Analyze the Augur's key features to build a trustless and efficient prediction market?
- B) You plan to introduce a custom token based on Ethereum. This token will be used within the app for rewarding contributions, unlocking premium features, and promoting posts. (5)
Based on the scenario, Analyze the key characteristics of application coins to design an effective token economy within your platform?

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